Lecture 16, Math 441

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Review for Midterm 1

$$\Psi(x, y) = x^2 y - \sin(y) x^2 + x^3$$

$$M = \Psi_x = 2xy - 2\sin(y)x + 3x^2$$

$$N = \Psi_y = x^2 - \cos(y)x^2$$

$$\left(2xy - 2\sin(y)x + 3x^2\right)\mathrm{d}x + \left(x^2 - \cos(y)x^2\right)\mathrm{d}y = 0$$

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Ex: $(2y - 2\sin(y) + 3x)\,\mathrm{d}x + (x - \cos(y)x)\,\mathrm{d}y = 0$

$M(x, y) = 2y - 2\sin(y) + 3x;\ N(x, y) = x - \cos(y)x$

$M_y \stackrel{?}{=} N_x$

$M_y = 2 - 2\cos(y)$ and $N_x = 1 - \cos(y)$

$M_y \neq N_x$, and the equation is not exact.

Integrating factor $\mu(x, y)$:

$\mu\,(2y - 2\sin(y) + 3x)\,\mathrm{d}x + \mu\,(x - \cos(y)x)\,\mathrm{d}y = 0$

$M = \mu\left(2y - 2\sin\left(y\right) + 3x\right)$ and $N = \mu\left(x - \cos\left(y\right)x\right)$

Set $M_y = N_x$:

$\mu_y\left(2y - 2\sin\left(y\right) + 3x\right) + \mu\left(2 - 2\cos\left(y\right)\right) =$

$\mu_x\left(x - \cos\left(y\right)x\right) + \mu\left(1 - \cos\left(y\right)\right)$

Assume $\mu_y = 0$:

$2\mu - 2\mu\cos\left(y\right) = \mu_x\left(x - \cos\left(y\right)x\right) + \mu - \mu\cos\left(y\right)$

$\mu - \mu\cos\left(y\right) = \mu_x\left(x - \cos\left(y\right)x\right)$

$\mu\left(\cancel{1 - \cos\left(y\right)}\right) = \mu_x\left(\cancel{1 - \cos\left(y\right)}\right)x$

$\mu = \mu_x x \qquad \frac{\mathrm{d}\mu}{\mu} = \frac{1}{x}\mathrm{d}x \iff \ln\mu = \ln x + A \iff \mu = Cx$

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$M = 2xy - 2x\sin(y) + 3x^2$ and $N = x^2 - x^2\cos(y)$

$$\Psi = \int M\,\mathrm{d}x = \int \left(2yx - 2\sin(y)x + 3x^2\right)\mathrm{d}x =$$

$$yx^2 - \sin(y)x^2 + x^3 + h(y)$$

$$\Psi_y = x^2 - \cos(y)x^2 + h'(y) = x^2 - x^2\cos(y)$$

So $h' = 0 \;\rightarrow\; h = \text{constant} = 0$

$$\Psi(x, y) = A$$

$$\boxed{yx^2 - \sin(y)x^2 + x^3 = A}$$

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$\frac{1}{y} = 2x \qquad \int \frac{1}{y}\mathrm{d}y = \int 2x\mathrm{d}x$

$\ln(y) = x^2 + A \qquad \boxed{y = C\mathrm{e}^{x^2}}$

$|y| = C\mathrm{e}^{x^2}$

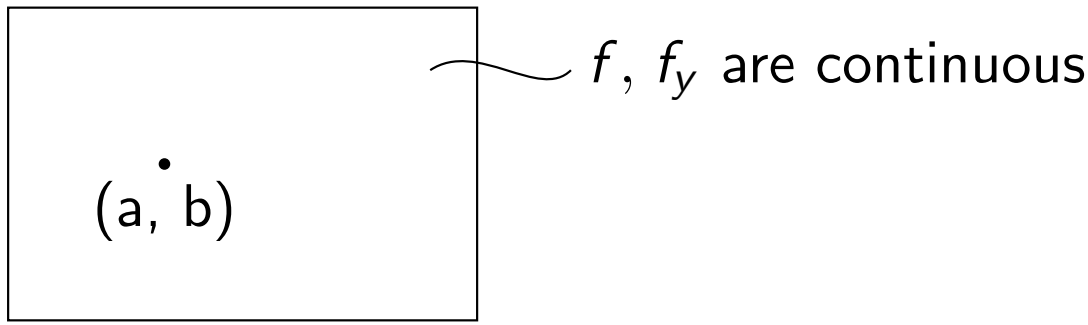
IC, $y(0) = +1 \qquad y(0) = C = 1 \qquad \boxed{y = \mathrm{e}^{x^2}}$

IC, $y(0) = -1 \qquad y(0) = C = -1 \qquad \boxed{y = -\mathrm{e}^{x^2}}$

1st Order IVP

$$\begin{cases} y' = f(x, y) \\ y(a) = b \end{cases}$$

If there exists a rectangle R

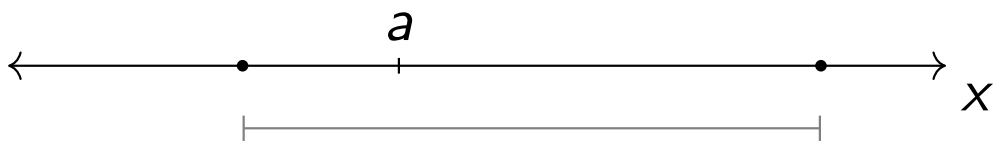


then there's 1 and only 1 solution somewhere around $x = a$.

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1st Order Linear IVP

$$\begin{cases} y' + p(x)y = q(x) \\ y(a) = b \end{cases}$$



I (on which p, q are continuous)

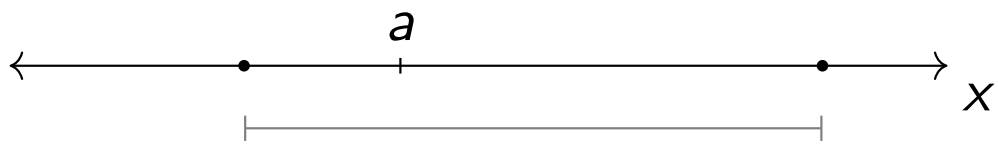
If $\exists$ an interval I around $x = a$ where p and q are continuous, then

$\exists$ one and only one solution over all of I.

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2nd Order Linear IVP

$$\begin{cases} y'' + p(x)\,y' + q(x)\,y = f(x) \\ \\ y(a) = b; \qquad y'(a) = c \end{cases}$$



I (on which p, q, f are continuous)

If $\exists$ an interval I around $x = a$ where p, q, f are continuous, then $\exists$ one and only one solution to the IVP over I.

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Extra conditions:

same point	different points
initial conditions	boundary conditions

$$\begin{cases} y'' + y = 0 \\ y(0) = 1 \\ y'(0) = 0 \end{cases}$$

$y = \cos(x)$

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$$\begin{cases} y'' + y = 0 \\ y(0) = 0 \\ y'(\frac{\pi}{2}) = 0 \end{cases}$$

$y = A\sin(x)$

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Singular Solutions

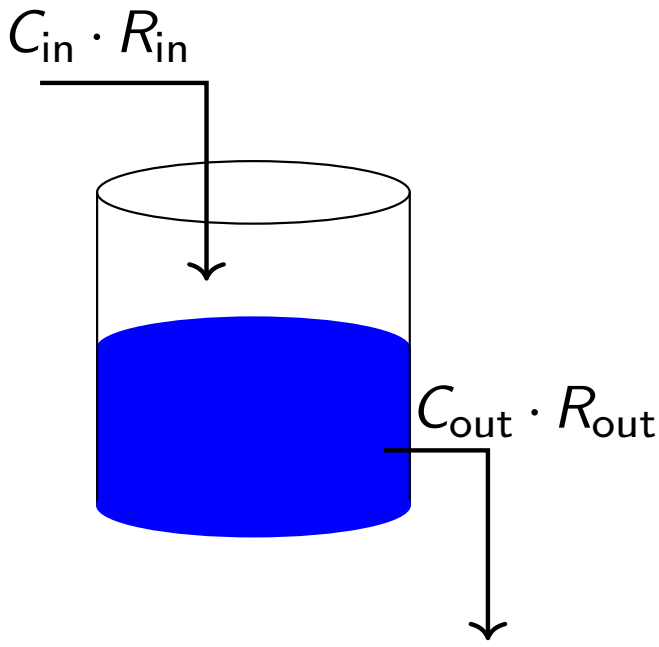
Ex: $y' = \frac{y^2}{x+1}$ (assuming $y \neq 0$)

$$\frac{dy}{y^2} = \frac{dx}{x+1} \rightarrow -\frac{1}{y} = \ln(x+1) + A \rightarrow$$

$$\boxed{y = \frac{1}{A - \ln(x+1)}} \quad \text{(general solution)}$$

But the function $\boxed{y(x) = 0}$ also solves the equation and it is not included in the general solution. It is a singular solution.

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(assume $R_{\text{in}} = R_{\text{out}}$ and $C = C_{\text{out}}$)

$Q = C \cdot V \qquad \frac{\mathrm{d}Q}{\mathrm{d}t} = V\frac{\mathrm{d}C}{\mathrm{d}t} = C_{\text{in}}R - CR$

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Analytic Solution of a Population Model

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = P^2 - 3P + 2 = (P-1)(P-2)$

Assuming $P \neq 1$ and $P \neq 2$, separate and use partial fractions

$$\frac{1}{(P-1)(P-2)} = \frac{A}{P-1} + \frac{B}{P-2} = \frac{A(P-2)+B(P-1)}{(P-1)(P-2)} = \frac{-2A-B+P(A+B)}{(P-1)(P-2)}$$

$P^0 = 1 = -2A - B \qquad P^1 = 0 = A + B$

$A = -B \qquad 2B - B = 1$

$B = 1 \qquad A = -1$

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$$\int \left(\frac{1}{P-2} - \frac{1}{P-1} \right) \mathrm{d}P = \int \mathrm{d}t$$

$$\ln(P-2) - \ln(P-1) = t + A$$

$$\ln\left(\frac{P-2}{P-1} \right) = t + A$$

$$\frac{P-2}{P-1} = C\mathrm{e}^{t} \qquad (C = \mathrm{e}^{A})$$

$$P - 2 = (P-1)C\mathrm{e}^{t} = PC\mathrm{e}^{t} - C\mathrm{e}^{t}$$

$$P - PC\mathrm{e}^{t} = 2 - C\mathrm{e}^{t} = P(1 - C\mathrm{e}^{t}) = 2 - C\mathrm{e}^{t}$$

Dividing, we get the explicit general solution:

$$\boxed{P(t) = \frac{2 - C\,\mathrm{e}^t}{1 - C\,\mathrm{e}^t}}$$

To this solution we need to add the singular solutions

$$\boxed{P(t) = 1; \qquad P(t) = 2}$$

after checking that they satisfy the equation.

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